Gordon Heltzel, PhD

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Overview

PhD-trained Quantitative UX Researcher who pairs rigorous measurement (psychometrics, logs & telemetry modeling, A/B testing) with AI-native workflows to evaluate AI-powered navigation experiences (Gemini, Google Maps) and a multi-national OEM partner ecosystem at Google.

- Lead quantitative research end to end: survey programs (HaTS, CSAT & sentiment benchmarking, user segmentation), behavioral logs and telemetry, and A/B tests that shape product roadmaps and technical specs.
- Advanced R, Python, and SQL to query, clean, and model large datasets, from behavioral log corpora ($N > 300{,}000$) to international survey panels, using regression, linear mixed-effects models, SEM, and IRT.
- Early adopter of LLM-assisted research workflows (prompt & context engineering, agent orchestration); built automated pipelines that cut research synthesis time 8x and upskilled UXRs and cross-functional peers.

Skills

Quantitative & Survey Methods: In-product experience measurement (HaTS, CSAT & sentiment benchmarking, user segmentation), Survey design & psychometrics, Longitudinal survey programs, Behavioral logs & telemetry analysis, A/B testing, Key Driver Analysis, Competitive benchmarking, Causal inference, Response-bias mitigation & statistical weighting

Statistical Techniques: Regression (linear mixed-effects, multivariate, logistic), SEM & Path modeling, Item Response Theory (IRT), PCA, EFA & CFA, Latent Profile Analysis & clustering, Kano modeling, Time-series modeling (DRT)

Technical Stack & Tools: R (advanced), Python, SQL; Colab, APIs, R Shiny dashboards; NLP & automated data pipelines; LLM-assisted workflows (Gemini; prompt & context engineering, agent orchestration); AI response evaluation

Experience

UX Researcher (Quantitative Specialist), Contract April 2025 – October 2026
Google, Android Auto *San Francisco Bay Area, CA*

- Lead end-to-end quantitative research for Android Auto's Gemini assistant and Google Maps navigation experiences, integrating HaTS and in-product surveys, logs and telemetry analysis, and A/B tests to evaluate agentic AI features across critical user journeys
- Partner with UXD, PMs, Data Scientists, and Eng to define, validate, and track UX metrics and key results, translating analyses into briefs, R Shiny dashboards, and decision frameworks that shape Eng/Product backlogs
- Build automated R/LLM research pipelines (prompt & context engineering, agent orchestration) and AI-developed taxonomies and rubrics that classify and score open-text data, cutting synthesis time 8x

Selected Projects:

- *Agentic AI Benchmarking & Response Evaluation*: Benchmarked Gemini's in-car agentic workflows against the leading competitor across navigation and routing (Google Maps) and other CUJs, measuring response quality on latency, accuracy, tone, safety, and conciseness. Applied regression, SEM, and Kano modeling to isolate CSAT drivers and built a bug-prioritization model that shaped Eng and Product backlogs.
- *AI Safety & DRT Telemetry Analysis*: Quantified driver cognitive load and safety of in-car LLM interfaces by triangulating Detection Response Task (DRT) log telemetry with A/B survey metrics. Applied linear mixed-effects models to isolate latency and usability effects, informing product design and technical specifications.
- *OEM Partner Ecosystem CSAT*: Ran the annual multi-national driver CSAT ($n \approx 1{,}800$) across seven car brands in Google's OEM partner ecosystem. Used Key Driver Analysis (regression) and statistical weighting to isolate satisfaction drivers, translating findings into roadmaps for OEM partner directors and internal PMs.
- *Psychometric KPI Overhaul*: Diagnosed flaws in legacy UX metrics; ran 16 cognitive interviews that cut redundant items from an expensive survey and seeded an SEM/IRT validation plan for a standardized sentiment benchmark across core CUJs.
- *HaTS & Longitudinal Teamfood Programs*: Analyzed HaTS intercept surveys on two digital car key implementations; recommended passive-entry unlock over phone-to-car unlock on comfort and reliability scores, and flagged latency and unlock-failure fixes for Eng. Modeled a longitudinal internal teamfood survey of technical users (Eng, PMs) to track experience quality over time.

Freelance Quantitative UX Researcher January 2025 – April 2025
Client: The Joint Chiropractic *Remote, USA*

Project Description: Designed custom survey instruments and modeled complex data in R. Applied Latent Profile Analysis to uncover data-driven user segments and isolate friction points, translating statistical findings into low-friction product changes that improved tooling efficiency and decreased bug and support reports by 64%.

Principal Researcher (Doctoral) Sep. 2017 – Jan. 2025
University of British Columbia *Vancouver, BC*

- Autonomously managed up to 6 concurrent empirical studies spanning digital behavior and user attitudes, applying a heavy quantitative toolkit to drive an internationally recognized research agenda
- Designed and fielded large-scale international surveys, lab experiments, and A/B tests, translating complex behavioral data into high-signal insights and interactive R Shiny dashboards
- Mentored junior researchers on survey construction, data manipulation, and robust experimental design

Selected Projects:

- *Behavioral Logs & API Data Pipelines*: Built API ingestion and NLP text-classification pipelines in R and Python, with custom codebooks to classify behavioral logs from Twitter/X ($N > 300{,}000$). Joined log telemetry with survey panels and human ratings to model online engagement and inform algorithmic content-ranking strategies.
- *Large-Scale Survey Programs & Longitudinal Measurement*: Designed and validated psychometric scales (IRT) and deployed longitudinal surveys across international panels ($n = 3{,}000$) to baseline sentiment and track divergence between user attitudes and behavior over time.

UX Research Consultant Feb. 2023 – May 2023
Client: Social Identity Lab *Vancouver, BC*

Project Description: Led foundational and secondary desk research for a digital platform. Conducted affinity mapping to translate user friction points into prioritized design recommendations, generating clear insights that shaped the core product roadmap for users facing systemic barriers in STEM.

Education

PhD, Social & Personality Psychology Sep. 2017 – May 2024
University of British Columbia *Vancouver, BC*

- Conducted high-impact research in a globally top-ranked, highly selective (4% acceptance rate) doctoral program
- **One of three dissertations globally** to receive the 2023 *Psychology of Technology Institute Dissertation Award*

BS, Psychological & Brain Sciences (Summa Cum Laude) Sep. 2013 – May 2017
Indiana University *Bloomington, IN*